

Name: _____



Checking and securing understanding of non-linear sequences

Task A: Arithmetic and geometric sequences

1) Categorise these sequences into arithmetic and geometric.

a) $-4, -2, 0, 2, \dots$

b) $2, 4, 8, 16, \dots$

c) $-4, 4, 12, 20, \dots$

d) $4, 12, 36, 108, \dots$

e) $36, 24, 12, 0, \dots$

f) $108, 36, 12, 4, \dots$

g) $1, -2, 4, -8, \dots$

h) $1, -2, -5, -8, \dots$

2) Find the blanked out 1st and 4th terms of these sequences.

a) The arithmetic sequence $\boxed{}, 9, 27, \boxed{}, \dots$

b) The geometric sequence $\boxed{}, 9, 27, \boxed{}, \dots$

c) The arithmetic sequence $\boxed{}, 4, 1, \boxed{}, \dots$

d) The geometric sequence $\boxed{}, 4, 1, \boxed{}, \dots$

Task B: Other types of sequences

1) Identify which of these sequences have a common second difference, and which have a common third difference.

a) 1, 2, 12, 31, 59, ...

b) 1, 2, 5, 11, 21, ...

c) 100, 95, 86, 65, 24, ...

d) -68, -66, -54, -32, 0, ...

2) Given that these sequences have a common second difference find the missing terms.

a) 1, 5, 11, , ...

b) , 23, 39, 62, ...

c) 20, 36, 48, ...

d) $\frac{1}{4}$, $\frac{1}{2}$, $\frac{5}{4}$, , , ...



3) Jacob is examining these three sequences.

a) 12, 20, 28, 36, 44, ...

b) 12, 20, 30, 42, 56, ...

c) 56, 70, 82, 92, 100, ...

I see three increasing sequences whereby the terms will always go up.



Jacob's statement is not entirely true. Explore the sequences and see if you can find the contradiction.